

Banner Tool

Architecture Documentation

Comprehensive System Design & Component Overview

1. System Overview

The Banner Tool is a professional React-based animated banner designer built for creating compliant pharmaceutical and healthcare marketing materials. It provides a comprehensive visual design interface with advanced animation capabilities, multi-artboard support, and specialized compliance features.

Key Capabilities:

- Multi-artboard design for multiple banner sizes simultaneously
- Advanced keyframe-based animation system with 300+ presets
- ISI (Important Safety Information) scroll components
- Full undo/redo history with state snapshots
- Real-time preview with GSAP timeline control
- Multiple element types: text, shapes, images, videos, HTML
- Export and packaging functionality
- Template-based workflows for EMR and animated modes

2. Architecture Layers

Layer	Components	Responsibility
Entry Point	main.tsx, App.tsx, ModeSelect	Application initialization
State Management	Zustand Store, designStore.ts	Global state and mutations
UI Components	MainLayout, Panels, Toolbar	User interface
Canvas System	Konva, DesignCanvas, Renderer	Element rendering
Animation Engine	GSAP, Keyframes, AnimationHelpers	Timeline control
Data Models	DesignElement, Artboard, Keyframe	Core data structures
Utilities	keyframes.ts, animations.ts	Helper functions
Build Tools	Vite, TypeScript, Vitest	Dev infrastructure

3. Component Hierarchy

The component structure follows a hierarchical composition pattern with `MainLayout` as the central orchestrator coordinating multiple specialized panels:

```
App
  ■■■ MainLayout (Central Orchestrator)
    ■■■ Toolbar (Tool Selection)
    ■■■ DesignCanvas (Konva Stage)
      ■ ■■■ Multiple Artboards
        ■ ■■■ ElementShape (per element)
        ■ ■■■ ISIScroll & ISIOverlay
        ■ ■■■ Transformer (selection)
    ■■■ PropertiesPanel (Edit properties)
    ■■■ Timeline (Animation editor)
    ■■■ LayersPanel (Layer management)
    ■■■ TemplatesPanel (Template library)
    ■■■ AssetsPanel (Media browser)
    ■■■ VariationsPanel (Size variations)
```

4. State Management (Zustand Store)

Elements Management

- elements array
- selectedId tracking
- Element CRUD operations

Animation System

- playheadTime
- isPlaying state
- Keyframe management

Artboard Management

- artboards array
- activeArtboardId
- Multi-view support

Canvas Properties

- canvasWidth/Height
- Background settings
- Visual properties

History

- past/future stacks
- undo()/redo() methods
- State snapshots

5. Animation Pipeline

Animation Flow:

1. User defines animation via Timeline or Properties panel
2. Animation preset or keyframes stored in DesignElement
3. Timeline triggers animation playback
4. AnimationHelpers.buildMasterTimeline() compiles keyframes
5. GSAP creates timeline with easing and interpolation
6. Each frame update triggers Konva canvas re-render
7. Canvas displays animated elements in real-time
8. User can scrub playhead to preview any frame

Animation Types:

- Keyframe Animations - Manual, frame-accurate control
- Preset Animations - 300+ from Animista.net catalog
- Entrance/Exit Animations - Automatic timing
- Timed Animation Blocks - Complex sequences
- Hover Effects - Interactive animations
- Easing Functions - 13+ preset curves

6. Element Types Supported

Text: Typography with font control, alignment, shadows, full animation

Rect: Rectangles with corner radius, stroke, fill, opacity, transforms

Circle: Circular shapes with radius control and animation

Image: Raster images with transform and animation support

Video: Video elements with properties similar to images

Shape: SVG Path-based shapes with fill/stroke

ISI Scroll: Healthcare compliance scrollable content with auto-scroll

HTML: Iframe wrapper for custom HTML content

7. Technology Stack

Category	Technology	Purpose
UI Framework	React 19.2	Component-based interface
State Mgmt	Zustand 5.0	Global state with subscriptions
Canvas	Konva 10.0	2D canvas drawing
Animation	GSAP 3.14	Professional animation timelines
Styling	Tailwind CSS 3.4	Utility-first CSS
Build Tool	Vite 7.2	Fast dev server & builds
Language	TypeScript 5.9	Type-safe JavaScript
Testing	Vitest 4.1	Unit & integration tests
Linting	ESLint 9.3	Code quality checks
Icons	Lucide React	Icon library